Quantum-in-the-Loop VLAM Alignment and Certified Control Safety

for in-vehicle and robot-controller compute — not a cloud VLAM

Tracks: RL Alignment (primary scored) · Safety (secondary scored) · Context: VLAM stages on ISO 26262 / IEC 62061 controllers, ≤100 ms inference
Public repository: https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026

Executive summary

–35.4%

training rollout budget vs the named GRPO baseline (3 seeds, mean ± SD) — and **+9.3 points** safer driving at equal budget, replicated on an independent machine; curriculum ≈ 2× budget-equivalence

78→100 %

episodes safe when the uncertified learned driver is wrapped by our certified runtime-assurance monitor — 22 µs per decision, with the safety envelope computed **before** measurement and confirmed inside the predicted interval on two machines

2 VLAM stages

quantum curriculum selector in **RL alignment** of the action policy; tensor-network Lyapunov wrap on the **control-output** stage — plus a **measured collective disturbance class** certifying the envelope, a validation input no classical pipeline produces. The 7B backbone stays a swappable black box — the deployment posture an OEM can consume

Why the numbers can be trusted

Every claim in this report carries a pre-registration frozen before execution, ≥3 seeds with mean ± SD, in-job controls, and cloud job IDs (~250 receipted quantum jobs across IBM Heron and QCi Dirac-3). Key results are replicated on a second, independent machine. Mandatory ablation isolates the quantum / QI component on both scored tracks (six-arm RL ladder; rank sweep on the certificate). A reviewer replicates every headline number from archived data with `pip install numpy && python verify.py` — no credentials required.

The same framework that placed as a GIC 2026 dual-track finalist (Mitsubishi/AIST) is why the VLAM-loop quantum stage is a controlled instrument: flow-native encoding (–26%, 3.5× tighter), predicted T2 (7–9% on 18 edges), inverted energy readout (E0 recovered to 0.006). Phase II feeds the certified safety envelope a derived jam-relaxation spectrum and couples it to Volkswagen's traffic stack — still train-time quantum, still ≤100 ms on the vehicle.

Technical Report

0. Challenge demand → this package

VLAMs at 7B–10B are the planned backbone of autonomous driving and robotics, and they do not fit in-vehicle or robot-controller compute without a training, compression, or safety story. This entry puts a **quantum-enhanced / quantum-inspired component into two VLAM stages** that decide whether that backbone can ship: **RL alignment of the action policy** (sample cost) and **formal assurance of the control output** (ISO 26262 / IEC 62061). The 7B weights stay a swappable black box — that is the OEM requirement, not a dodge.

Challenge demand	This package
VLAM stage (required)	(1) post-training **RL alignment** of the action head; (2) **runtime safety** on the control-output stage
Bottleneck — model footprint 7B–10B	Bar already met classically (INT4: 3.97× @ 2.45%, our measurement, §7) — quantum effort concentrated where headroom exists
Bottleneck — attention $O(N^2)$	Phase II roadmap (annex)
Bottleneck — RL alignment cost	–**35.4%** rollouts vs named GRPO baseline (bar ≥10%); **+9.3 pt** safer at equal budget
Bottleneck — control safety	Uncertified driver **77.9%** → **100%** safe; envelope sealed *before* measurement; **22 µs** (4,500× under 100 ms)
≥3 runs, mean ± SD; ablation isolating Q/QI	3 seeds everywhere; six-arm RL ladder + rank ablation (§2–§4)
4–8 page report; public repo + README	This document + https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026

Claim (§)	Evidence artifact	Receipt
RL: 35.4% under GRPO budget, 3/3 seeds (§2)	`vw_rl_{A,B,C,D,E}_s{21,22,23}.json`	71 Dirac-3 job IDs in JSONs
RL: +9.3-pt robustness at equal budget; ~2× budget-equivalence, two-box (§2)	`rl_finalquality_classical.json` + `RL_FINALQUALITY_BOX.md`	replicated, independent machine
Safety: 78→100% under RTA; envelope predicted then confirmed (§3)	`vw_rta_demo.json`, `vw_lyap_bias_sweep.json`	3 seeds × 500 episodes ×2 machines
Flow encoding of the RL quantum stage: –26%, 3.5× tighter (§5)	`flow_encoding_test.json`	12 paired Dirac-3 jobs, frozen prediction
Device-layer control of that quantum stage: ζ, T2, inverted E0 (§5)	`ncomp_regrade_*`, `t2_bridge_result.json`, `npoint2_result_*`	fez + kingston, banked
Supporting: S(q,ω) jam-relaxation, two-sided classical boundary (annex)	`vw_jam_relaxation_spectrum.json`, `A1_FINAL_VERDICT.md`	fez `d9rm4j9dsedc73agrb70`; kingston `d9rdfb1dsedc73agh5ng`

Every number in this report regenerates from archived receipts (§8).

1. VLAM integration — the component, not a cloud VLAM

The field's two named blockers, and why this architecture answers both. The 2026 VLAM-for-driving literature converges on the same pair of obstacles: inference latency of **500–2000 ms**, "far exceeding control cycle requirements" (DriveVLM-RL, arXiv:2603.18315; VLAM survey, arXiv:2512.16760), and susceptibility to **hallucination** — outputs "inconsistent with visual input." Against a 100–300 ms authorization budget, the first is a 5–20× structural overrun; the second is an uncertified action reaching the actuator.

These are not two problems but one deployment consequence: **a VLAM cannot be the thing that decides, so it must be wrapped by something certified and fast.** That is precisely what §3 delivers — a 22 µs certified runtime-assurance monitor (4,500× inside the budget) with a verified fallback, wrapping a learned policy that is treated as untrusted by construction. The wrap does not require the VLAM to become faster or stop hallucinating; it bounds what either failure can do. On this reading the safety track is not a compliance exercise but the component that makes VLAM deployment tractable at all — and the disturbance class it is certified against is computed on quantum hardware (§3), which is where our quantum contribution enters the critical path of the field's actual blocker.

A VLAM is perception → reasoning → **action**. The 7B–10B backbone is what OEMs cannot train or host on the vehicle. Two stages around that backbone are where a quantum / QI component can change the product without moving inference to a GPU cluster:

1. **Alignment (scored).** After (or instead of) expensive preference fine-tuning, the action policy is aligned by RL under domain randomization. We put a QCi Dirac-3 annealer *inside every GRPO iteration* as the curriculum selector. Quantum work is **train-time only**. 2. **Control safety (scored).** The VLAM's action is an uncertified command. A tensor-network Lyapunov monitor + verified fallback wraps that command at **22 μs**, 4,500× inside the ≤**100 ms** inference budget. Only monitor + fallback are certified; the VLAM stays replaceable without re-certification — the ISO 26262 / IEC 62061 posture an OEM can consume.

Plant for both tracks: kinematic-bicycle lane-keeping (lateral / heading error, $|\delta| \le 0.25$, v = 25 m/s) under crosswind and impulses — the **control substrate a VLAM commands**, not a 7B forward pass. Scope is declared: we do not claim a compressed 7B checkpoint or an $O(N^2)$ attention replacement. We claim the two stages that make a VLAM deployable on a vehicle or robot controller.

2. RL-alignment track (scored): quantum-in-the-loop training

Method. Each GRPO iteration submits the 64-candidate rollout pool (start × disturbance metadata) to a QCi Dirac-3 quantum annealer, which selects the 16-rollout curriculum (frontier + diversity objective) before any simulation is spent. Task: lane-keeping alignment under domain randomization (kinematic bicycle, v=25 m/s regime, crosswind ±1.2 rad/s, impulses); linear-Gaussian policy from zero initialization; target = within 5% of nominal-controller evaluation reward.

Budget result (rollouts-to-target, seeds 21/22/23):

arm	selection	mean ± SD
A — GRPO-32, random (**named baseline**)	586.7 ± 98.8	
B — random-16 (size control)	unreliable (1/3)	
C — Dirac-3-16, untuned	**378.7 ± 52.9 (−35.4%)**	
D — classical optimizer, same objective	fails 0/3 (curriculum collapse)	
E — classical Gibbs sampler, tuned T	298.7 ± 106.9 (2× the device's variance; needs per-task temperature tuning the device does not)	
C′ — device on optimizer-grade encoding	fails 0/3 (confirms D's mechanism)	

Product result (the stronger claim): at *equal* budget the curriculum produces a measurably safer driver. Held-out stress suite (bias 0.9–1.35 rad/s + impulses): curriculum-trained policies 76.4 ± 0.7% safe vs baseline 67.1 ± 1.3% (**+9.3 points**). Independently replicated on a second machine (+7.2 pts seeds 21–23; +18.4 pts seeds 21–25), with the budget–quality Pareto: **curriculum at half the budget matches or beats the baseline at every rung** (E@320 ≥ A@640; E@640 > A@1280 ≈ 2× budget equivalence), and a variance collapse (baseline SD 22–28 pts — seeds that never learn — vs curriculum SD 1.8): **curriculum selection de-risks the training run.** Robust across stress windows (+16 to +21 pts).

Attribution, stated exactly. The Pareto rungs quoted above are **arm E** (the tuned classical Gibbs curriculum), not arm C (the device). On the raw rollouts-to-target metric arm E is the stronger arm — **298.7 ± 106.9 versus the device's 378.7 ± 52.9** — so the device does not win that metric against every classical arm, and we do not claim it does. The device's measured advantages on this track are specific and narrower: it beats the **named GRPO baseline** by 35.4%, requires **zero tuned parameters** where arm E requires per-task temperature tuning, and carries **roughly half the seed variance**. §4 isolates the mechanism.

3. Safety track (scored): certified runtime assurance

Method. Tensor-network Lyapunov monitor $V(x)=||L\cdot\phi(x)||^2$ (3-level tensor-product features, rank 3 of 9), grid+Lipschitz certified, wrapping an uncertified learned policy (behavior-cloned MLP with covariate shift and 100 ms delay): policy drives; verified fallback engages at V > 0.75c with hysteresis. Only monitor + fallback face certification — the learned driver (stand-in for the VLAM action head) stays a black box, swappable without re-certification. The same wrap applies to a robot controller under IEC 62061: certified monitor, uncertified policy, deterministic fallback.

Results (3 seeds × 500 episodes/condition): policy alone **77.9 ± 25.0%** safe at the certified boundary → **100.0 ± 0.0%** wrapped; benign conditions: guard idles (1–12% intervention). Monitor overhead **22 μs/decision** (4,500× under the 100 ms budget). **Sealed-prediction envelope:** the safe disturbance margin was computed *before* measurement (holds to 1.0 rad/s; collapse predicted in [1.2, 1.6]) and the measured collapse landed inside the predicted interval — on two machines, under two perturbation protocols.

Versus the named heuristic baseline (tuned clipping, threshold sweep in receipts): the certificate delivers what no heuristic reaches at any tuning — the envelope is known *a priori* (a heuristic's margin is discoverable only by crashing); zero tuned parameters (trigger = computed c); and the formal $V>0$, $dV\leq 0$ evidence chain — the artifact an ISO 26262 / IEC 62061 case consumes.

Certification under the measured disturbance class (quantum input to the scored track). A certified envelope is only as good as the disturbance model it is validated against. We evaluated the certified plant under three disturbance ensembles at *identical rms power*: white noise, an AR(1) surrogate matched to variance and autocorrelation (the standard classical validation models), and the **measured collective disturbance spectrum** — the real-frequency jam-relaxation object computed on quantum hardware, full k=0–15 series on the 64-rung interacting lattice, dominant collective line 0.0625 cycles/step reproduced on two devices (fez `daa9pn4e74ec73akj9i0`; annex flight). Results (3 seeds × 1,000 episodes per cell): the certified boundary against the measured spectrum is **0.60 rad/s rms** (95.6 ± 0.8% safe; 98.4 ± 0.3% at 0.50). The two classical validation models miss it from opposite sides: **white-noise validation grades 100.0 ± 0.0% safe at every amplitude through 1.3 rad/s — where the true safe rate is 39%** — unbounded over-certification; the AR surrogate reads **25.2 ± 2.1% at the certified boundary itself**, a 70-point misjudgment at the operating point, and would force certification several-fold below the true envelope. Spectral depth is load-bearing: certifying against a shallow (16-point anchor-sector) spectrum places the boundary at 0.88 rad/s — a **32% envelope error that only the full-depth computation catches**. The vehicle-side story is unchanged: quantum computes the disturbance model at certification time; the car runs the 22 µs monitor. Receipts: `results/vw_cert_gap_score_v2.json`, `results/vw_cert_gap_boundary_v2.json`, `results/vw_jam_relaxation_spectrum_v2.json`.

4. Mandatory ablation — isolating the quantum / QI component

RL (six-arm ladder). The ladder isolates the winning mechanism: random selection fails; a deterministic classical optimizer of the identical objective breaks training outright (0/3, curriculum collapse); only near-optimal *stochastic* selection trains. The device delivers that ingredient natively — zero tuned parameters, the tightest seed variance of any arm, executing inside every iteration of the scored run (71 receipted jobs) — where the best classical alternative needs per-task temperature tuning and carries 2× the variance. Arm C′ closes the causal loop: pushing the device toward deterministic optimization reproduces the classical optimizer's failure — the component works *because of* its native sampling character.

Safety (rank axis). r=2 certifies nothing (expressivity floor); **r=3 interior-optimal, 0.462 ± 0.030 certified area**; full rank unstable (± 0.110 with a collapsed seed); in the saturation-bound regime the TN certificate covers **1.57 ± 0.29×** the best quadratic's area (all seeds ≥ 1.39). The optimal rank tracks plant dimension (r=3 at d=2, growing with d) — a measured design rule. Removing the low-rank tensor structure costs certified area *and* reliability.

5. Why the VLAM-loop quantum component is a controlled instrument

The RL stage is not a black-box annealer call. The same framework that placed as a GIC 2026 dual-track finalist predicted, before running, that encoding the curriculum QUBO on the annealer's native flow manifold (no penalty walls) beats naive binary encoding. Measured on the identical selection objective, 6 paired trials, 12 receipted jobs: **flow encoding** –**26% better objective, 3.5× tighter spread** — frozen prediction confirmed. That is the design rule behind the 71-job curriculum and the method any OEM optimization on analog quantum hardware should use.

The same devices that will carry Phase-II mobility spectra are now **operated as an instrument, not a hope.** Idle ZZ (ζ) is a spectrometer (two independent readouts; 9/9 kingston and 8/9 fez edges vs a 24-day atlas). Encoded T2 is predicted from that atlas ($\Gamma_{Bell} = \Gamma_a+\Gamma_b$; median error 7% / 9% on 18 edges) — a jam-relaxation or curriculum tile's coherence budget is a number *before* the first shot. Energy and spectra are visibility-inverted, not raw-summed: the native 64-cell operator recovers E0 on **5/5 live tiles** (best –2.523 vs –2.518, –**0.006**); filler edges with L0 < 0.90 were a layout error and are no longer graded. Computation sits in even-ZZ sectors the noise cannot see (logical X_L 9/9; GHZ-4 protection passed).

What this changes for Volkswagen. The scored RL number does not move — it is already receipted. Its *provenance* is now named: put the VLAM-loop problem on the machine's native manifold. The scored safety envelope stays the ISO artifact; Phase II feeds it a *derived* disturbance model (collective $S(q,\omega)$ of interacting traffic, read with inverted estimators) instead of a grid heuristic on a 2-state plant. That is full framework capability applied to the challenge's two VLAM stages, not a second supremacy claim.

Receipts (banked, no copy flights): `npoint2_result_20260830_014105.json`, `npoint2_result_20260829_182700.json`, `t2_bridge_result.json`, `ncomp_regrade_20260830_122917.json`, `ncomp_regrade_20260830_123936.json`, `nc1_energy_inverted.json`.

6. Deployment architecture

Quantum work is train/validation-time only — vehicle inference latency is the classical monitor: **22 μs**, 4,500× under the ≤**100 ms** requirement. The runtime-assurance pattern makes the uncertified VLAM swappable without re-certification. Every quantum output is classically verified before use: selected curricula are index sets; certificate candidates pass the deterministic grid+Lipschitz certifier. Quantum hardware is never a trusted in-vehicle component — the correct posture for an ISO 26262 / IEC 62061 pipeline.

7. Resource allocation — where quantum effort pays

A pre-registered study of the compression bottleneck established, with receipts, that INT4 quantization already meets the challenge bar classically (3.97× at 2.45% drop vs the ≥2× @ ≤5% requirement) and that the remaining mixed-precision allocation space is flat — no allocator, classical or quantum, has headroom left to harvest there. On the way to that finding the device solved its allocation objective *exactly* (DP gap 0.0, 9/9 instances) and beat random allocation 5–25× on real accuracy. The finding is itself a deliverable: it tells an OEM precisely where **not** to spend quantum budget, and it is why this entry concentrates quantum effort on the two bottlenecks with measured headroom — training cost (§2) and certified safety (§3). Attention $O(N^2)$ is Phase II scope (annex). The demonstrations run on a 2-state control substrate with compact policies; the 7B backbone remains a swappable black box by design (§1) — the deployment posture an OEM can consume.

7b. Related work and positioning

VLAM deployment. The blockers this entry targets are the ones the field names: inference latency of 500–2000 ms against a 100–300 ms control budget, and hallucinated actions inconsistent with visual input (DriveVLM-RL, arXiv:2603.18315; *VLAM for Autonomous Driving: Past, Present, Future*, arXiv:2512.16760). Recent work attacks these inside the model — AlphaDrive applies RL and reasoning to VLM planning, AutoVLA discretises trajectories into action tokens, ALN-P3 aligns perception, prediction and planning. Our position is complementary and deliberately outside the model: **treat the backbone as untrusted and swappable, and certify the wrap.** That is what makes the safety artifact survive a model upgrade without re-certification.

Quantum in automotive. Published automotive quantum work is predominantly annealing-based combinatorial optimisation — traffic routing (Volkswagen/D-Wave, Lisbon), materials and battery chemistry (Volkswagen/Google), and paint-shop scheduling. This entry is a different class: a quantum component *inside the RL training loop*, and a certified control artifact whose disturbance model is computed as a real-time many-body spectrum. To our knowledge no published automotive quantum work certifies a control envelope against a hardware-computed disturbance class.

Where we do not claim novelty. The runtime-assurance pattern (certified monitor plus verified fallback wrapping an uncertified policy) is established in aerospace and control literature. Our contributions are the quantum-computed disturbance class it is certified against, and the measurement that classical surrogate disturbance models misjudge the certified boundary in both directions (§3).

8. Reproducibility

Every result: a frozen pre-registration written before execution, raw data stored verbatim, ≥3 seeds with mean ± SD, in-job controls, and cloud job IDs. Key results replicated on a second, independent machine. All scripts and JSON receipts accompany this submission (§0 map); every hardware figure regenerates from archived counts with no credentials — `pip install numpy && python verify.py` replicates every headline number. Deadline: GIC 2026 Phase I closes 2026-09-15; this package is submitted with runway.

Annex. Supporting hardware record (not a scored track)

The two scored tracks are the challenge answer. This annex is why the mobility *disturbance class* those tracks will consume in Phase II is a real-frequency object, not an ill-posed continuation.

For interacting mobility systems the industrially decisive quantity is **S(q,ω), the jam-relaxation spectrum**. Imaginary-time plus analytic continuation is mathematically ill-posed at any size. Quantum hardware evolves on the real-time axis; the spectrum is a Fourier transform with **no continuation step**. Shipped: X(q,t) of a 156-qubit congestion interface on IBM Heron and `vw_jam_relaxation_spectrum.json` (dominant collective line 0.0625 cycles/step across q = 0.5–2.0).

Two-sided adjudication. Hostile classical attack on our own data (engines 1e-16 vs dense; joins 1e-15; all blocks twice): **k** ≤ **6** fully reproduced (0.4σ at k=4; five wavevectors within 1.4σ at k=6); **k** ≥ **8** the attack died (operators past 2×10^8 terms) and the referee's verdict stands: *"k≥8 cells stand contested with measured receipts."* Hardware cells there: **8.9**σ **and 11.5**σ, clean k=0 control. Exact-theorem anchors 0.9836 / 0.9806; derived 18-line spectrum beats a 20%-detuned control (R^2 0.87 vs 0.71). In-kind CHSH anchor **S = 2.3604, +35.9**σ, same-day replica +37.0σ.

These cells will not be re-flown as a copy. Phase II *re-reads* them under the §5 instrument (atlas layout, even-sector encoding, predicted T2, L0 invert) and couples the spectrum to the §3 envelope and Volkswagen's traffic stack.

Team Merlin Digital. One framework, one instrument: a quantum component in the VLAM alignment loop, a certified wrap on the VLAM action, and a measured disturbance class no classical pipeline produces — every number on the same receipt class, from cloud job to report table.